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Effect of Localized Rotation Errors on the Deutsch–Jozsa Algorithm

Submitted in Partial Fulfillment of the Requirements for the Internship Module

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Contents

1. Introduction

Quantum computing has emerged as a promising computational paradigm that exploits the principles of quantum mechanics to solve certain problems more efficiently than classical computers. Despite their theoretical advantages, current quantum computers face significant practical challenges. Qubits are highly sensitive to imperfections in quantum gate operations, environmental noise, and decoherence. These errors can alter the quantum state during computation, leading to incorrect measurement outcomes and reducing algorithmic performance. Consequently, understanding how errors affect quantum algorithms is essential for the development of reliable quantum computing systems. Identifying the stages of a quantum circuit that are most vulnerable to errors is particularly important for improving the robustness of quantum computations.

The Deutsch–Jozsa algorithm provides a suitable framework for investigating the effects of quantum errors because its circuit structure is relatively simple and its ideal output is known exactly. This makes it possible to clearly observe how imperfections influence algorithm performance. In this work, quantum circuit simulations were used to investigate the effect of localized single-qubit rotation errors introduced into the Deutsch–Jozsa circuit. The errors were applied at different locations within the algorithm to model imperfections that may occur during practical implementations of quantum hardware.

By systematically varying the error location and comparing the resulting algorithm performance, this study aims to identify the most sensitive stages of the Deutsch–Jozsa circuit. In addition, the influence of different rotation axes and rotation angles on the probability of obtaining the correct measurement outcome is examined. The results provide insight into error propagation within quantum circuits and contribute to a better understanding of the robustness of quantum algorithms in the presence of localized gate imperfections.

2. Theoretical Background

2.1 The Deutsch–Jozsa Algorithm

The Deutsch–Jozsa algorithm is one of the earliest quantum algorithms to demonstrate a computational advantage over classical methods. It is designed to determine whether a Boolean function is constant or balanced using a single evaluation of a quantum oracle. The algorithm exploits the principles of quantum superposition, interference, and parallelism. By preparing the input qubits in a superposition of all possible states, the oracle can evaluate the function for all inputs simultaneously. A subsequent interference step amplifies the information required to distinguish between constant and balanced functions, allowing the result to be obtained with a single oracle query [1].

The Deutsch–Jozsa problem considers a black-box quantum oracle that implements a Boolean function

$$f: \{0,1\}\ n \rightarrow \{0,1\}$$

The function takes n-bit binary values as input and produces either a 0 or a 1 as output for each such value. We are promised that the function is either constant (0 on all inputs or 1 on all inputs) or balanced (1 for exactly half of the input domain and 0 for the other half). The task then is to determine if f is constant or balanced by using the oracle.

For a deterministic classical algorithm, for a conventional deterministic algorithm where n is the number of bits, $2^{n-1} + 1$ evaluations of the function are required in the worst case. This is because observing identical outputs for exactly half of the possible inputs is insufficient to prove that the function is constant; one additional evaluation is needed to rule out the possibility that the function is balanced. In contrast, the Deutsch–Jozsa algorithm can determine whether the function is constant or balanced using a single oracle query, demonstrating a computational advantage over deterministic classical approaches [2].

2.2 Oracle Representation in the Deutsch–Jozsa Algorithm

To generate our quantum circuit to implement the Deutsch-Jozsa algorithm, we will use some of the same components as before:

1. The input and output registers are represented using Dirac notation $|x\rangle$.

2. Two quantum registers, $|x\rangle$ and $|y\rangle$, are used as inputs to the oracle U_f. The register $|x\rangle$ stores the input state, while $|y\rangle$ serves as an ancillary qubit. The ancillary qubit enables the oracle to encode the Boolean function $f(x)$ through the transformation $y \oplus f(x)$ while preserving the reversibility required for quantum operations.

3. The oracle produces two output registers. The first output register remains unchanged as $|x\rangle$, while the second output register is transformed according to the Boolean function $f(x)$ resulting in the state $|y \oplus f(x)\rangle$. and the other that is the **XOR** of the input register x and the input register x XORed with the function $f(x)$ as $|\ x, y \oplus f(x)\rangle$.

The complete oracle transformation can therefore be written as

$$U_f\colon |x\rangle, |y\rangle \rightarrow |x\rangle, |y \oplus f(x)\rangle$$

Since quantum operations are represented by unitary transformations, the oracle must be reversible. The inverse operation is given by

$$U_f^{-1}\colon |x, y \oplus f(x)\rangle \rightarrow |x, y\rangle$$

The reversible oracle representation described above forms the basis of the Deutsch–Jozsa circuit and is illustrated in ***Figure 2.2.1.***

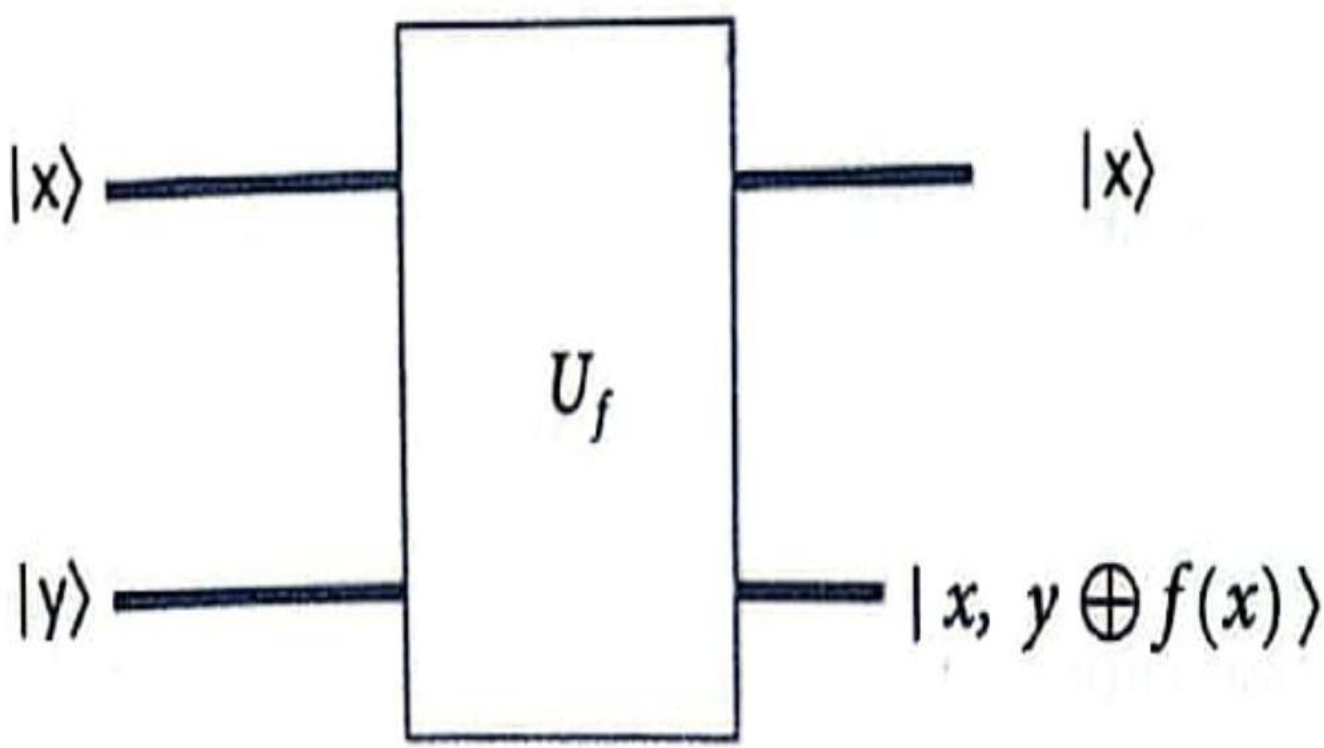

Figure 2.2.1*: Graphical representation of the oracle function* U_f [3].

2.3 Operation of the Deutsch–Jozsa Algorithm

For the Deutsch–Jozsa algorithm to operate correctly, the Boolean function $f(x)$ must be implemented as a quantum oracle. The oracle performs the transformation U_f while preserving the quantum superposition of the input states. Since quantum operations are represented by unitary transformations, the oracle must also be reversible. The specific implementation of the oracle is not important for the algorithm; it is only required to perform the prescribed transformation associated with the Boolean function $f(x)$.

- Register Initialization: Initialize an $(n + 1)$ qubit quantum register consisting of n input qubits and one ancilla qubit. The input register is prepared in the state $|0\rangle^{\otimes n}$, while the ancilla qubit is initialized in the state $|1\rangle$.
- Creation of Superposition: Apply Hadamard gates to all qubits, creating a uniform superposition of all possible computational basis states.

- Oracle Operation: Apply the oracle U_f, which evaluates the Boolean function $f(x)$ on all basis states simultaneously through quantum parallelism.

- Interference Stage: Apply Hadamard gates to the n input qubits. This operation causes constructive and destructive interference of the probability amplitudes, enabling the distinction between constant and balanced functions.
- Measurement: Measure the input register and determine the probability (P(0…0)) of obtaining the all-zero state$|0\rangle^{\otimes n}$. In the ideal Deutsch–Jozsa algorithm, P (0…0)=1, for constant functions and P(0…0)=0, for balanced functions. This probability serves as the primary performance metric throughout this study and is used to quantify the impact of localized rotation errors on the Deutsch–Jozsa algorithm [1].

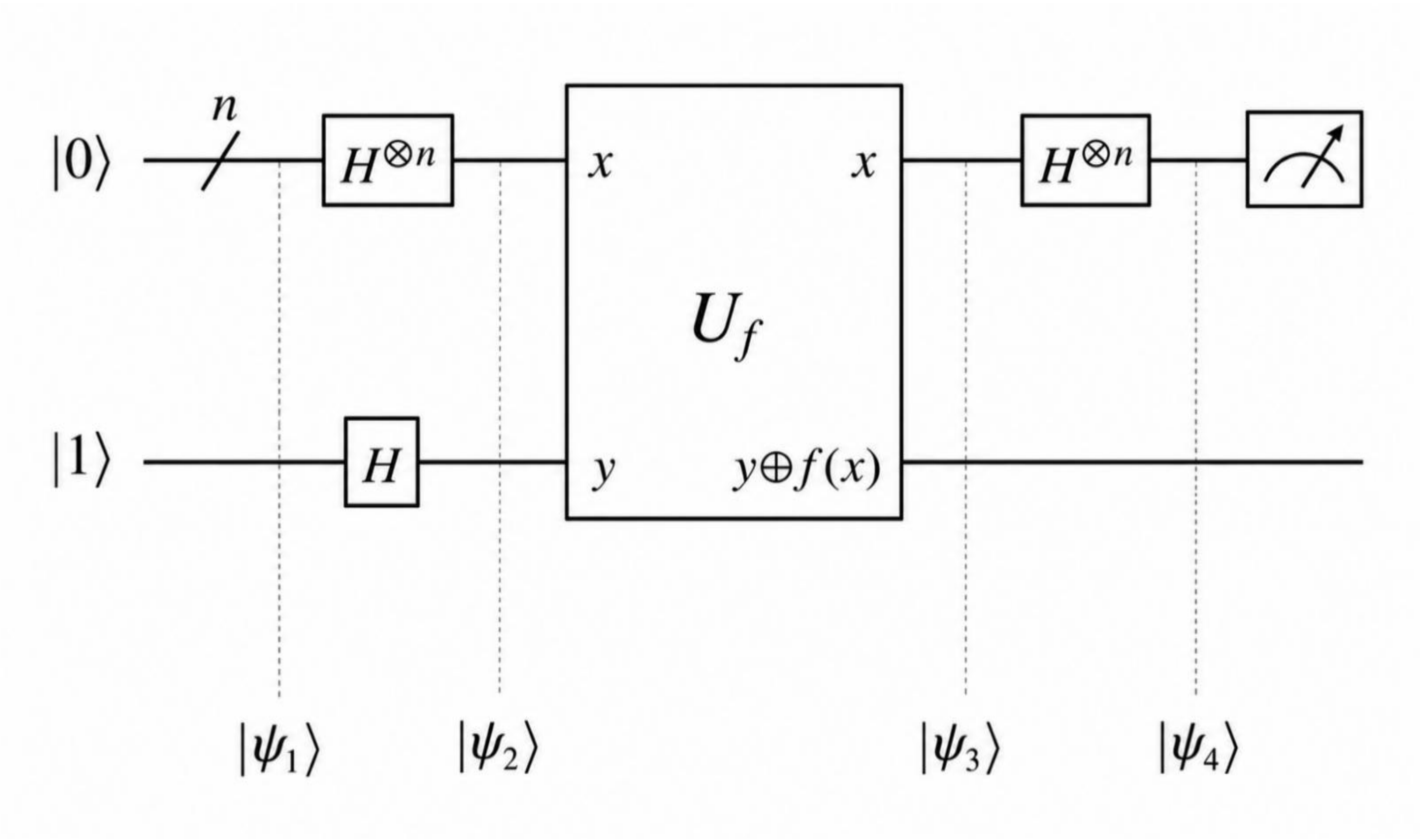

Figure 2.3.1: *Deutsch-Jozsa algorithm quantum circuit* [2].

The algorithm begins with the $n + 1$ bit state $|0\rangle^{\otimes n}|1\rangle$ That is, the first n bits are each in the state $|0\rangle$ and the final bit is $|1\rangle$. A Hadamard gate is applied to each bit to obtain the state

$$\frac{1}{\sqrt{2^{n+1}}}\sum_{x=0}^{2^n-1}|x\rangle\,(|0\rangle-|1\rangle)$$

where x runs over all n -bit strings, which each may be represented by a number from 0 to $2^n - 1$. We have the function f implemented as a quantum oracle. The oracle maps its input state $|x\rangle|y\rangle$ to $|x\rangle|y \oplus f(x)\rangle$, where $\oplus$ denotes addition modulo 2. Applying the quantum oracle gives

$$\frac{1}{\sqrt{2^{n+1}}}\sum_{x=0}^{2^n-1}|x\rangle(|0 \oplus f(x)\rangle - |1 \oplus f(x)\rangle)$$

For each x, $f(x)$ is either 0 or 1. Considering the two possible values of $f(x)\rangle$ the state can be rewritten as

$$\frac{1}{\sqrt{2^{n+1}}}\sum_{x=0}^{2^n-1}(-1)^{f(x)}|x\rangle(|0\rangle-|1\rangle)$$

Since the ancillary qubit ($\frac{|0\rangle-|1\rangle}{\sqrt{2}}$) is separable from the input register, it can be omitted from the subsequent analysis

$$\frac{1}{\sqrt{2^{n}}}\sum_{x=0}^{2^n-1}(-1)^{f(x)}|x\rangle$$

Applying a Hadamard transformation to each input qubit yields. The total transformation over all n qubits can be expressed with the following identity:

$$H^{\otimes n}|k\rangle = \frac{1}{\sqrt{2^{n}}}\sum_{j=0}^{2^n-1}(-1)^{k\cdot j}|j\rangle$$

$where\ j \cdot k = j_0k_0 \oplus j_1k_1 \oplus \cdots \oplus j_{n-1}k_{n-1}$ denotes the modulo-2 inner product of the binary strings j and k. Applying this identity gives

$$\frac{1}{\sqrt{2^n}}\sum_{x=0}^{2^n-1}(-1)^{f(x)}\left[\frac{1}{\sqrt{2^n}}\sum_{y=0}^{2^n-1}(-1)^{x\cdot y}\left|y\right\rangle\right] = \sum_{y=0}^{2^n-1}\left[\frac{1}{2^n}\sum_{x=0}^{2^n-1}(-1)^{f(x)}(-1)^{x\cdot y}\right]|y\rangle$$

From this, we can see that the probability for a state k to be measured is

$$\left|\frac{1}{2^n}\sum_{x=0}^{2^n-1}(-1)^{f(x)}(-1)^{x\cdot k}\right|^2$$

The probability of measuring k = 0, corresponding to $|0\rangle^{\otimes n}$, is

$$\left|\frac{1}{2^n}\sum_{x=0}^{2^n-1}(-1)^{f(x)}\right|^2$$

which evaluates to 1 if $f(x)$ is constant (constructive interference) and 0 if $f(x)$ is balanced (destructive interference). In other words, the final measurement will be $|0\rangle^{\otimes n}$ (all zeros) if and only if $f(x)$ is constant and will yield some other state if $f(x)$ is balanced [2].

3. Methodology

3.1 Software and Development Environment

The project was implemented in Python using a collection of scientific and computing software tools. Numerical calculations, matrix operations, and data processing were performed using NumPy. JupyterLab used for interactive notebook environment facilitated testing, debugging, and data analysis throughout the project. Version control was managed using Git, enabling systematic tracking of code development and modifications during the internship.

3.2 Error Model Implementation

To investigate the effect of errors on the algorithm, single-qubit rotation errors were introduced at different locations within the quantum circuit, as shown in Figure 3.1.,

Four error locations were considered: E_1, E_2, E_3, and E_4. The error at each location was modelled as a single-qubit rotation acting on a target input qubit. Localized errors were modelled using the rotation operators $R_x(\theta)$, $R_y(\theta)$, and $R_z(\theta)$, where θ denotes the rotation angle and determines the magnitude of the introduced error. These operators represent rotations about the X-, Y-, and Z-axes of the Bloch sphere, respectively. This arrangement enables the sensitivity of different stages of the Deutsch–Jozsa algorithm to localized noise to be evaluated systematically.

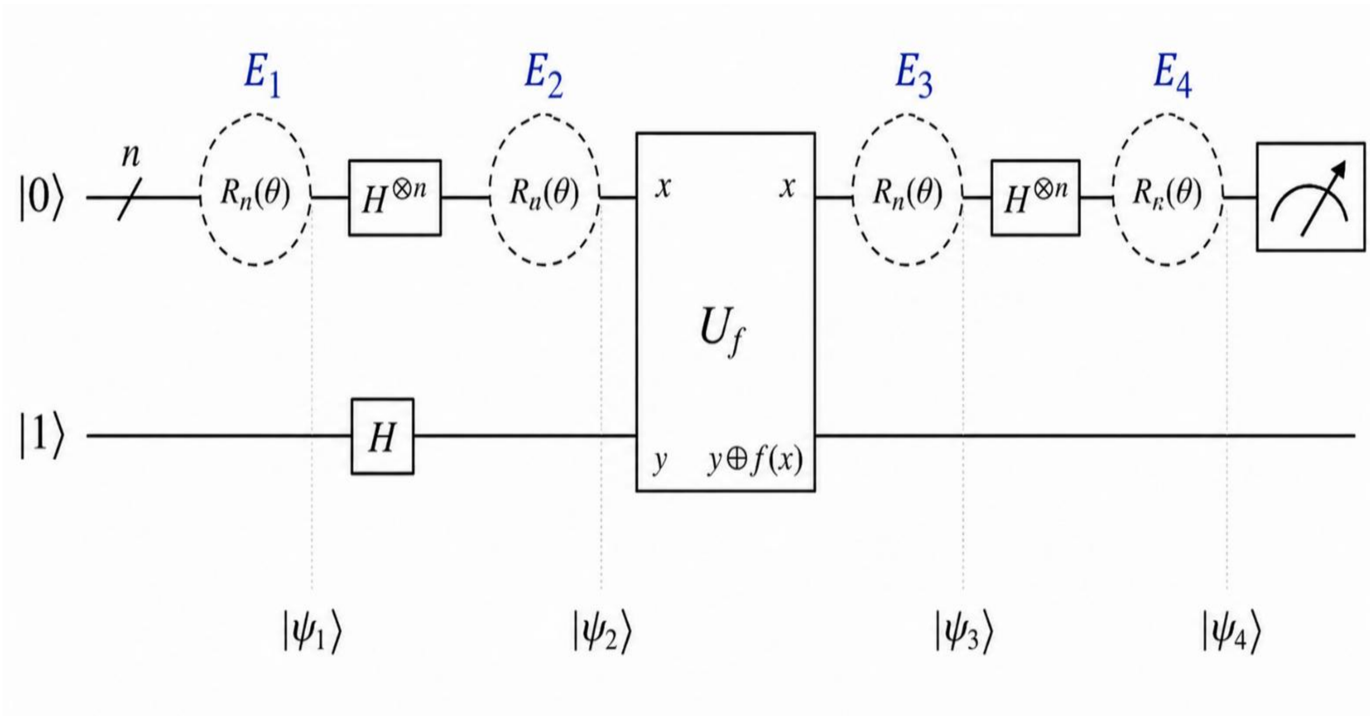

Figure 3.1: *Modified Deutsch–Jozsa circuit within localized error with axis of* $R_n(\theta)$.

The physical interpretation of the four error locations is summarized as follows:

- $\boldsymbol{E_1}$ – Error introduced before the first Hadamard gate. This error affects the preparation of the initial quantum state.
- $\boldsymbol{E_2}$ – Error introduced after the first Hadamard gate and before the oracle U_f . This error affects the superposition state used as the input to the oracle.
- $\boldsymbol{E_3}$ – Error introduced after the oracle and before the final Hadamard gate. This error affects the quantum state immediately after the oracle has encoded the function information.
- $\boldsymbol{E_4}$ – Error introduced after the final Hadamard gate and before measurement. This error affects the final state and may directly influence the measurement outcome.

These four locations correspond to the state preparation, oracle input, oracle output, and measurement stages of the algorithm, allowing the sensitivity of each stage to localized noise to be investigated separately.

3.3 Simulation Parameters

The rotation error was applied to the target qubits, from input qubits. This choice provides a consistent and unbiased error location across different system sizes.

The simulations were performed for (n = 1, 3, 5, 7,) and (9) input qubits. The probability using 1024 measurement shots for each circuit execution. Preliminary tests using fewer shots exhibited larger statistical fluctuations, whereas 1024 shots provided more stable probability estimates while maintaining reasonable computational efficiency. Rotation angles were varied from (0°) to (180°) in increments of (10°), Localized errors were implemented using the rotation operators allowing the effects of X-, Y-, and Z-axis rotation errors to be investigated systematically.

4. Results and Discussion

The algorithm's performance is assessed using the probability of obtaining the correct measurement outcome. Deviations from this probability indicate performance degradation due to errors.

4.1 Effect of Error Position under X-Axis Rotation Errors, $\boldsymbol{R_x}(\boldsymbol{\theta})$

In the ***Figure 4.1.1***, The observed sensitivity at E_1 and E_4suggests that the state-preparation and measurement stages are the most vulnerable to X-axis rotation errors. The probability decreases steadily as the rotation angle increases, near zero at 180°. The algorithm is particularly sensitive to disturbances occurring during state preparation and the final measurement stage.

In contrast, error at $E_{2,}$ and E_3 have almost no observable effect. Both curves remain close to the ideal result, therefore overlap with the ideal curve in the plot. This behavior can be explained by the fact that the qubits are predominantly in the Hadamard-basis states $|+\rangle$ and $|-\rangle$ at these locations. Since these states are eigenstates of the Pauli X operator, X axis rotations primarily introduce phase factors rather than changing the measurement probabilities.

Overall, For X-axis rotations, the state preparation and measurement stages are the most vulnerable locations.

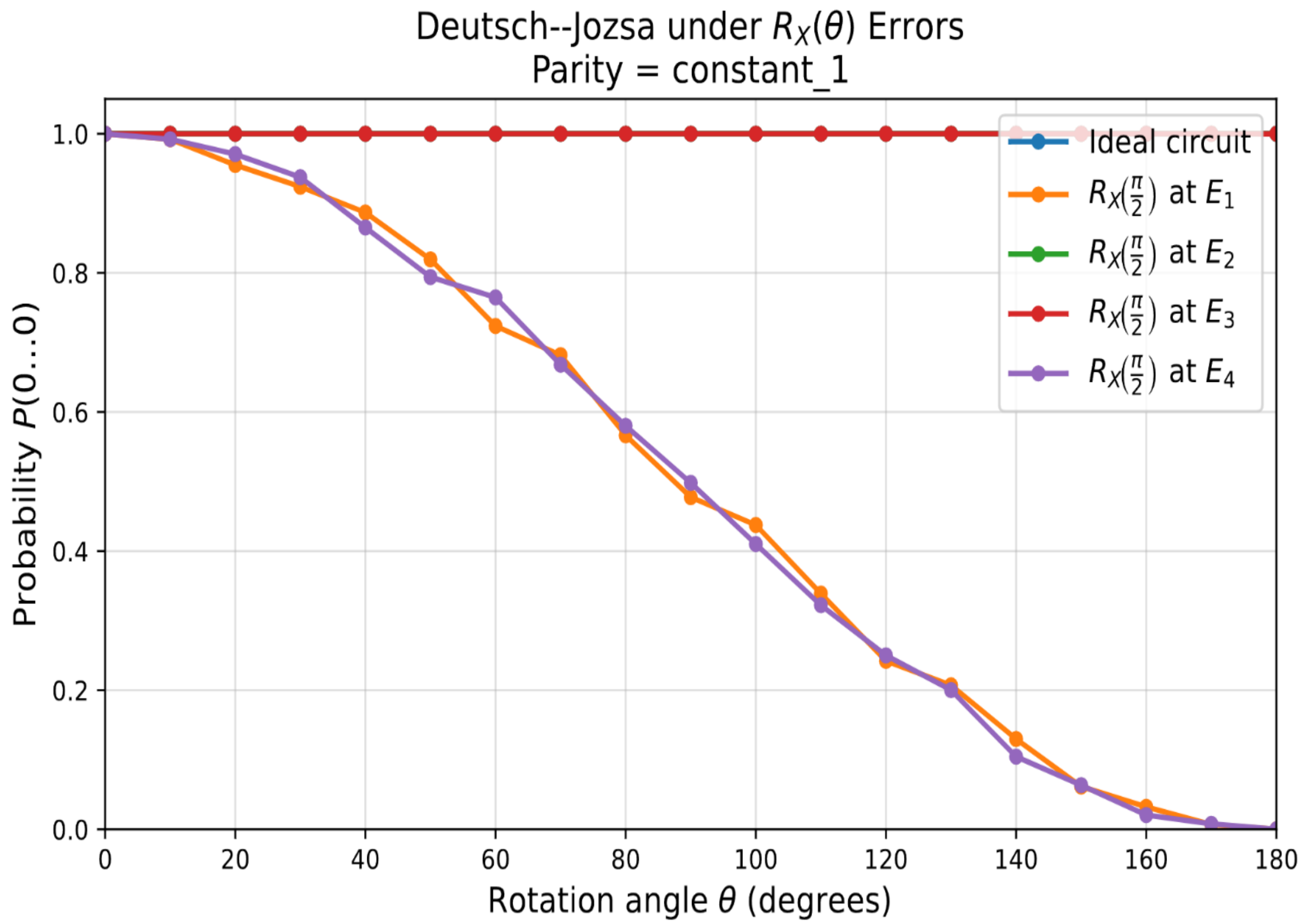

Figure 4.1.1 *Effect of X-axis rotation errors,* $R_x(\theta)$ *, on the performance of the algorithm.*

4.2 Effect of Error Position under Y-Axis Rotation Errors, $R_y(\theta)$

In ***Figure 4.2.1***, All four-error exhibit very similar behavior. This result indicates that the algorithm is similarly sensitive at Y-axis rotation errors in the whole circuit. From this we see that the angle of rotations has a greater impact than the position of the error in the circuit.

Overall, the results demonstrate that Y-axis rotation errors affect the Deutsch–Jozsa algorithm uniformly across all stages of the circuit.

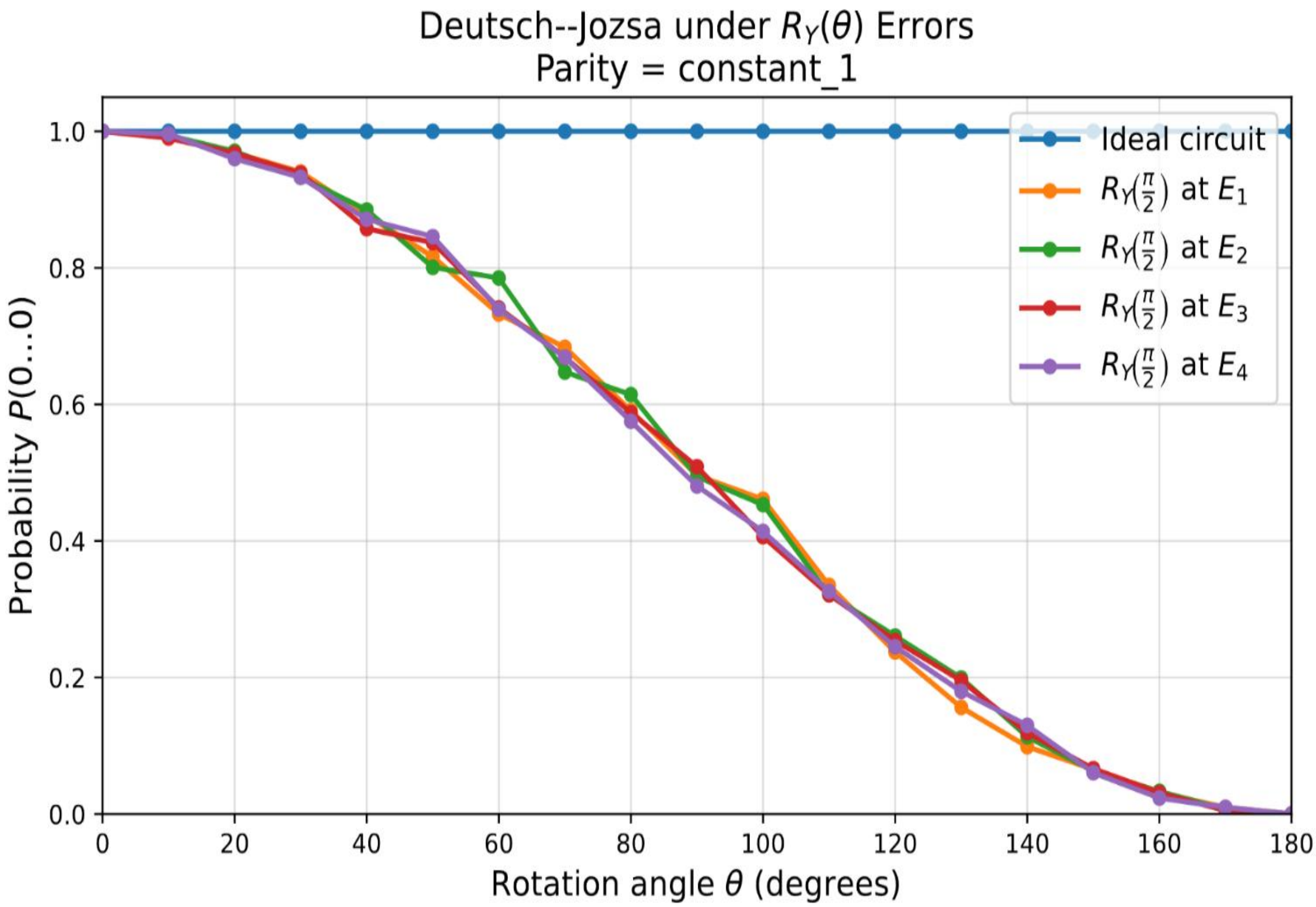

Figure 4.2.1 *Effect of Y-axis rotation errors on the performance on the algorithm.*

4.3 Effect of Error Position under Z-Axis Rotations, $R_z(\theta)$

In ***Figure 4.3.1,*** E_1,and E_4 have almost no observable effect on the final measurement outcome. The corresponding curves remain essentially identical to the ideal circuit.

This behavior can be explained by the nature of Z-axis rotations, which primarily introduce phase shifts. Before state preparation and immediately before measurement, these phase shifts do not significantly affect the measured computational basis probabilities. However, when the errors occur during the intermediate stages of the circuit, particularly after the creation of superposition states, they modify the relative phases responsible for quantum interference. As a result, the algorithm becomes highly sensitive at E_2, and E_3.

The results demonstrate that phase errors have the greatest impact when they occur within the interference generating region of the Deutsch–Jozsa circuit. This finding contrasts with the X axis case, where the largest degradation was observed at E_1, and E_4.

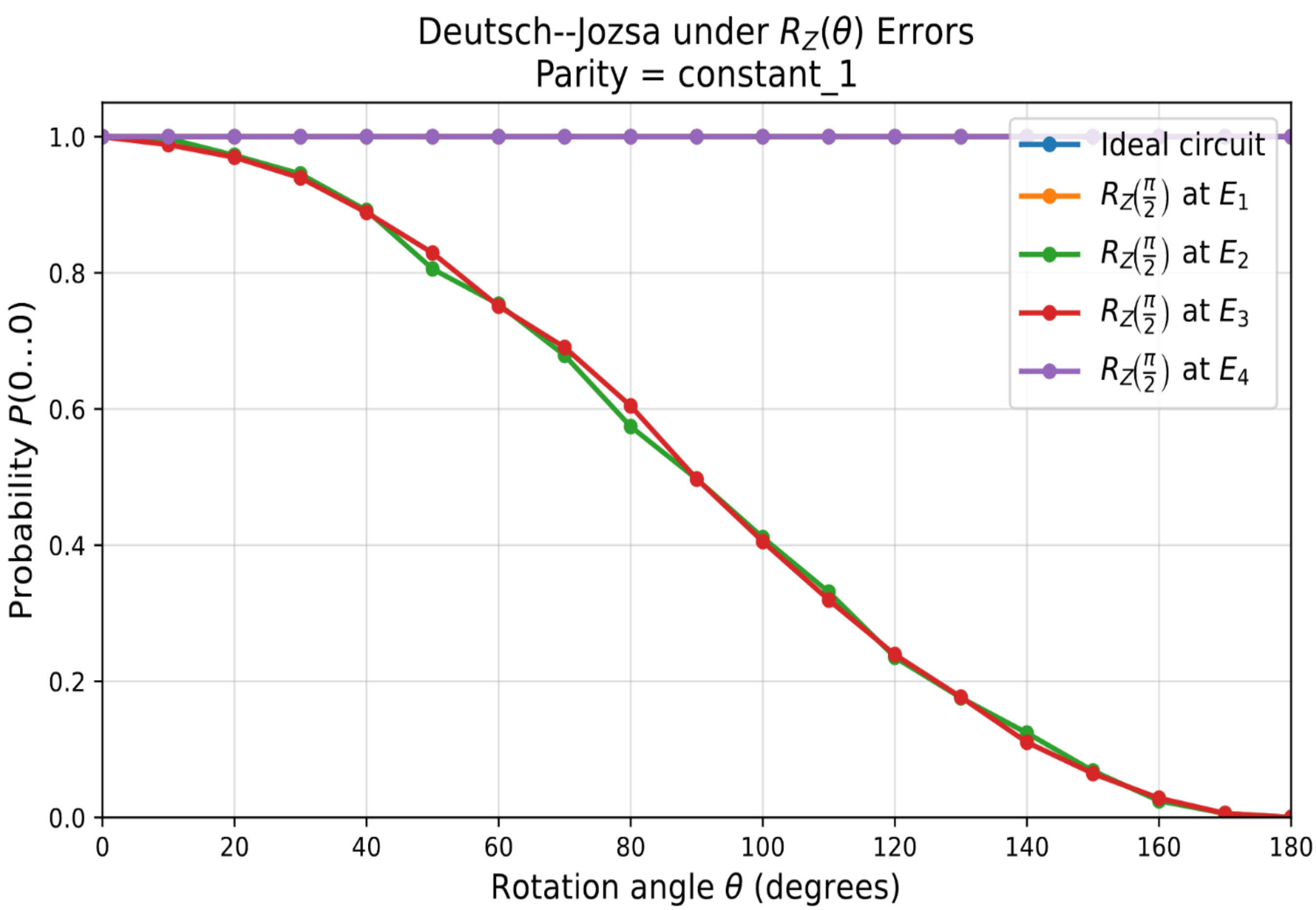

Figure 4.3.1*, The effect of Z-axis rotation errors on the performance of the.*

Note: The results for **Constant_0** ($f(x) = 0$) are not shown here because they are identical to those obtained for **Constant_1** ($f(x) = 1$)

4.4 Results for the Balanced Function

In ***Figure 4.4.1***, shows the effect of rotation errors on the Deutsch–Jozsa algorithm for a balanced function. The probability of measuring the all-zero state. remained approximately zero for all investigated rotation angles, error locations, and rotation axes. As a result, all curves overlap throughout the entire range of rotation angles, the rotation errors do not produce an observable change in the measurement.

Since identical behavior was observed for X-, Y-, and Z-axis rotation errors, only the X-axis results are presented here as a representative example.

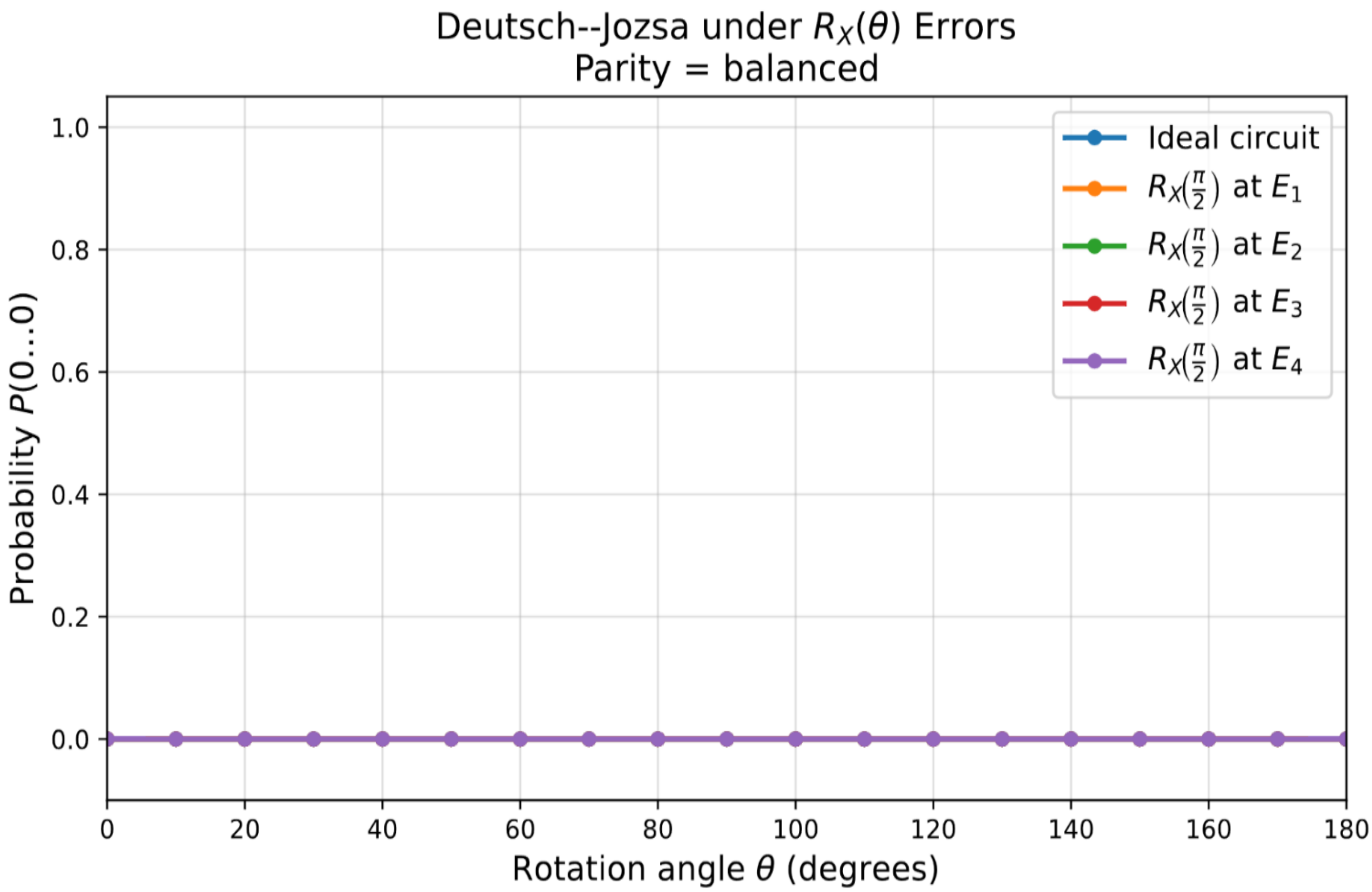

Figure 4.4.1*, The effect of X-axis rotation errors on the performance for a balanced function.*

Note: Only the results for X-axis rotation errors are shown for the balanced function, as the corresponding results for Y-axis and Z-axis rotation errors exhibit the same behavior.

4.5 Comparison of Error Locations

Sensitivity metrics provide a quantitative measure of how strongly localized rotation errors affect the performance of the Deutsch–Jozsa algorithm. By comparing the ideal and error-affected probabilities, they enable the identification of the most influential error locations and rotation axes. This provides an objective assessment of algorithm robustness and highlights the regions that would benefit most from error mitigation.

n	**Function**	**Axis**	**Error Location**	**Maximum sensitivity**	**Mean Sensitivity**
9	constant_1	X	E1	1.0	0.501
9	constant_1	X	E2	0.0	0.000
9	constant_1	X	E3	0.0	0.000
9	constant_1	X	E4	1.0	0.497
9	constant_1	Y	E1	1.0	0.499
9	constant_1	Y	E2	1.0	0.501
9	constant_1	Y	E3	1.0	0.500
9	constant_1	Y	E4	1.0	0.501
9	constant_1	Z	E1	0.0	0.000
9	constant_1	Z	E2	1.0	0.501
9	constant_1	Z	E3	1.0	0.497
9	constant_1	Z	E4	0.0	0.000

Table 4.5.1: Maximum and Mean Sensitivity Values for constant functions

The high maximum sensitivity values 1.0 observed for all Y-axis error has significant influence on algorithm performance. For X-axis is confined to E_1, and E_4, while the Z-axis exhibits sensitivity only at E_2, and E_3. Mean sensitivity values close to 0.5 for sensitive cases further confirm a sustained response rather than isolated peak effects. Consequently, sensitivity analysis provides an effective mechanism for identifying influential error sources and assessing the robustness of the algorithm under localized errors.

The sensitivity analysis supports the three observations. Maximum sensitivity values of 1.0 were obtained for all Y-axis error, for X-axis errors only at E_1 and E_4, and for Z-axis errors only at E_2 and E_3.

The sensitivity profiles shown in ***Appendix A (Figures A.1–A.3)*** visually support the quantitative results presented in ***Table 4.5.1.*** Together, they provide complementary quantitative and qualitative assessments of error sensitivity across the X-, Y-, and Z-axis rotations.

4.6 Effect of Depolarizing Noise on the Deutsch–Jozsa Algorithm

The depolarizing channel is one of the most widely used noise models in quantum information theory. In this model, a qubit remains unchanged with probability $1 - p$, and is depolarized with probability p, causing the quantum state to lose information and move toward a maximally mixed state [4].

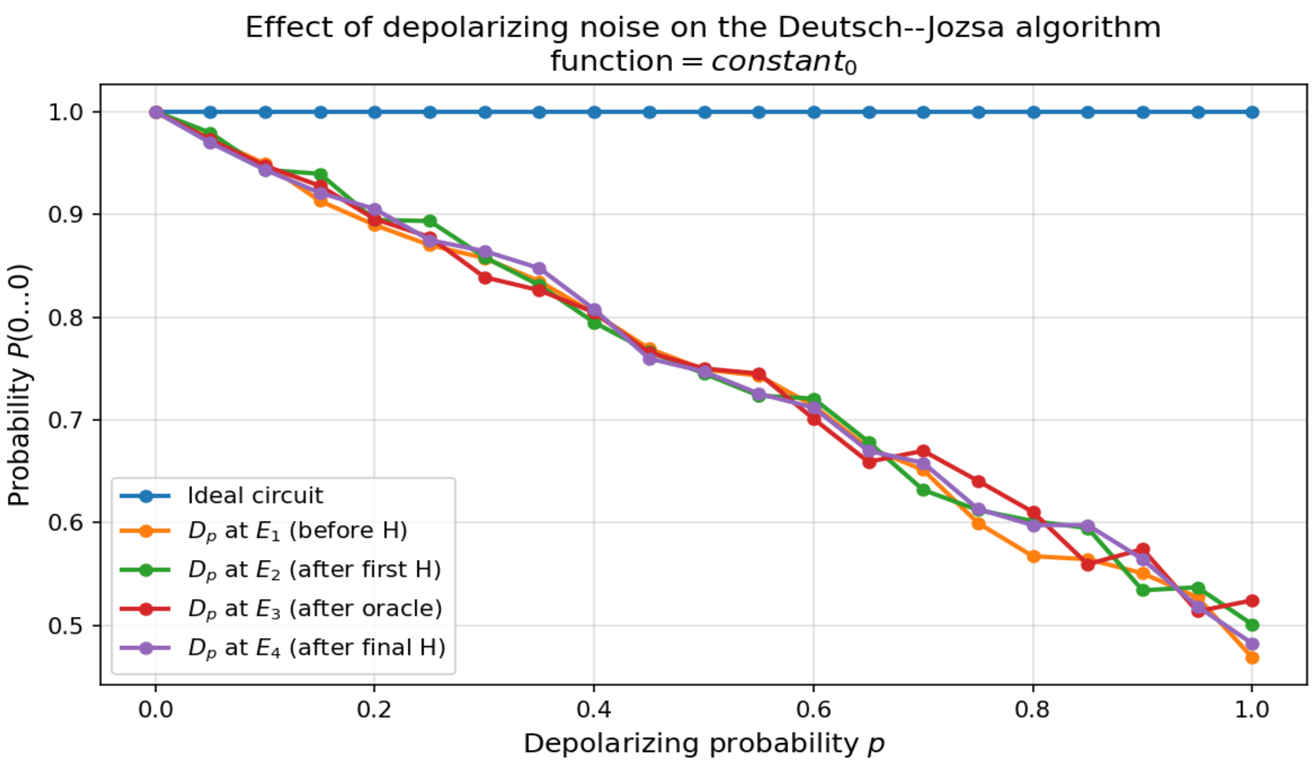

Figure 4.6.1, illustrates the effect of a depolarizing channel on the Deutsch–Jozsa algorithm.

Depolarizing channel was introduced at the four error locations E_1, E_2, E_3, and E_4. shows that the probability of obtaining the correct output decreases approximately linearly as the depolarizing probability p increases. the depolarizing channel produced nearly identical results at all four circuit locations. This indicates that the algorithm is primarily sensitive to the overall noise strength rather than the specific location where the noise is introduced. Consequently, depolarizing noise causes a uniform loss of quantum information throughout the circuit, resulting in a location-independent reduction in the probability of obtaining the correct output.

Note that: The fundamental difference between coherent and stochastic errors. Rotation errors correspond to coherent unitary perturbations whose impact depends on both the error type and the stage of the computation, and depolarizing channel represents stochastic quantum noise that randomly degrades both amplitude and phase information, causing a uniform loss of quantum information throughout the circuit.

5. Conclusion

This study investigated the effect of localized rotation errors and depolarizing noise on the performance of the Deutsch–Jozsa algorithm. The results demonstrate that the impact of an error depends strongly on the underlying error model. While localized rotation errors are sensitive to both the type of rotation and its location within the circuit, depolarizing noise depends primarily on the overall noise strength.

- For X-axis rotation errors, the strongest degradation in performance was observed at error locations E_1 and E_4, corresponding to the state-preparation and measurement stages of the circuit. In contrast, Z-axis rotation errors had the greatest effect at E_2 and E_3, where quantum interference plays a central role. For Y-axis rotation errors, all four error locations exhibited similar behavior, indicating that the rotation angle has a greater influence on performance than the specific error location.
- Depolarizing noise produced a nearly uniform reduction in algorithm performance across all circuit locations. Unlike coherent rotation errors, depolarizing noise acts as a stochastic process that randomly degrades quantum information, resulting in a location-independent loss of performance.

Overall, these findings identify the most sensitive stages of the Deutsch–Jozsa circuit under different error models and provide insight into the propagation of errors in quantum algorithms. The results contribute to a better understanding of quantum circuit robustness and may support the development of more effective error-mitigation strategies for future quantum computing systems.

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Appendix A: Sensitivity Analysis

The sensitivity metric, $S(\theta) = |P_{\text{error}} - P_{\text{ideal}}|$

was evaluated for each rotation axis, where P_{error} denotes the probability obtained in the presence of a rotation error and P_{ideal} is the corresponding ideal probability. The resulting sensitivity curves are presented below [5].

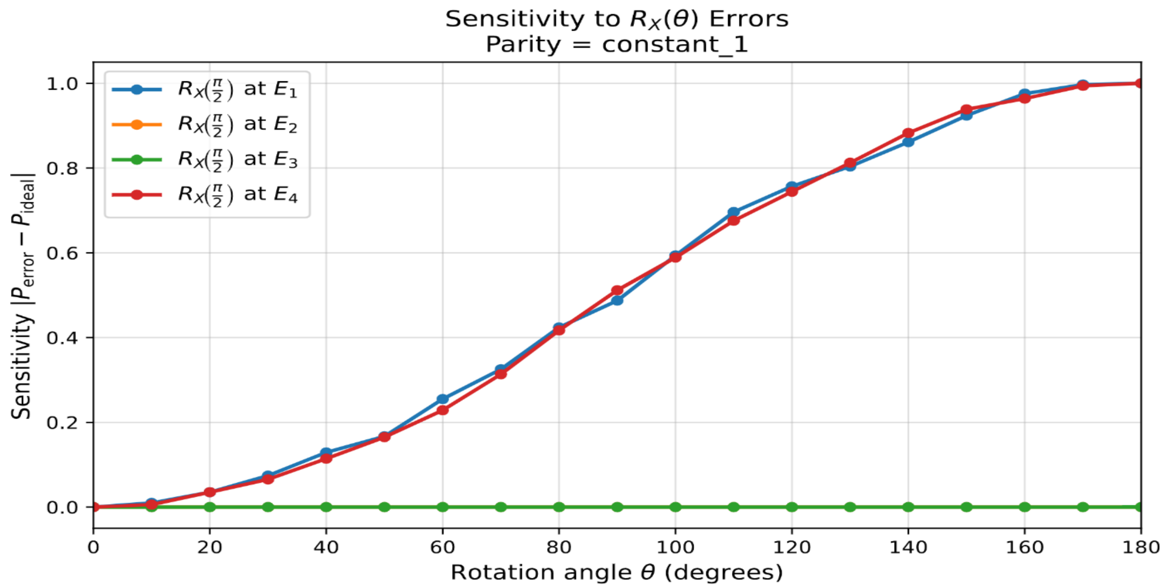

Figure A.1: Sensitivity to $\boldsymbol{R_x(\theta)}$ errors for the Deutsch–Jozsa algorithm.

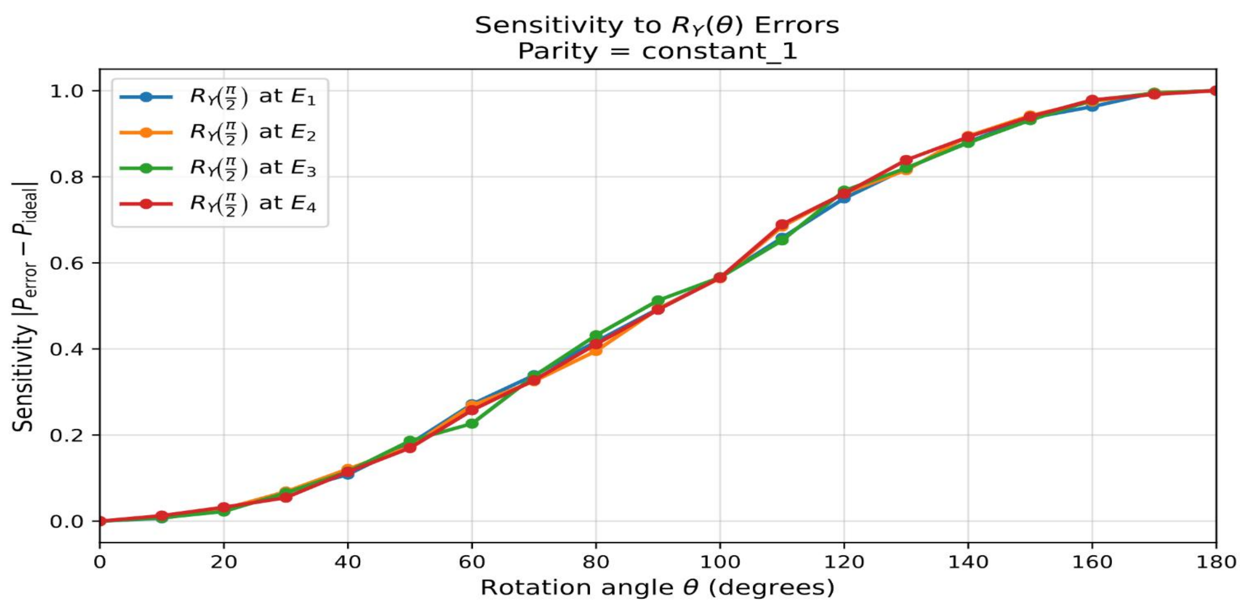

***Figure A.2*:** Sensitivity to $\boldsymbol{R_y(\theta)}$ errors for the Deutsch–Jozsa algorithm.

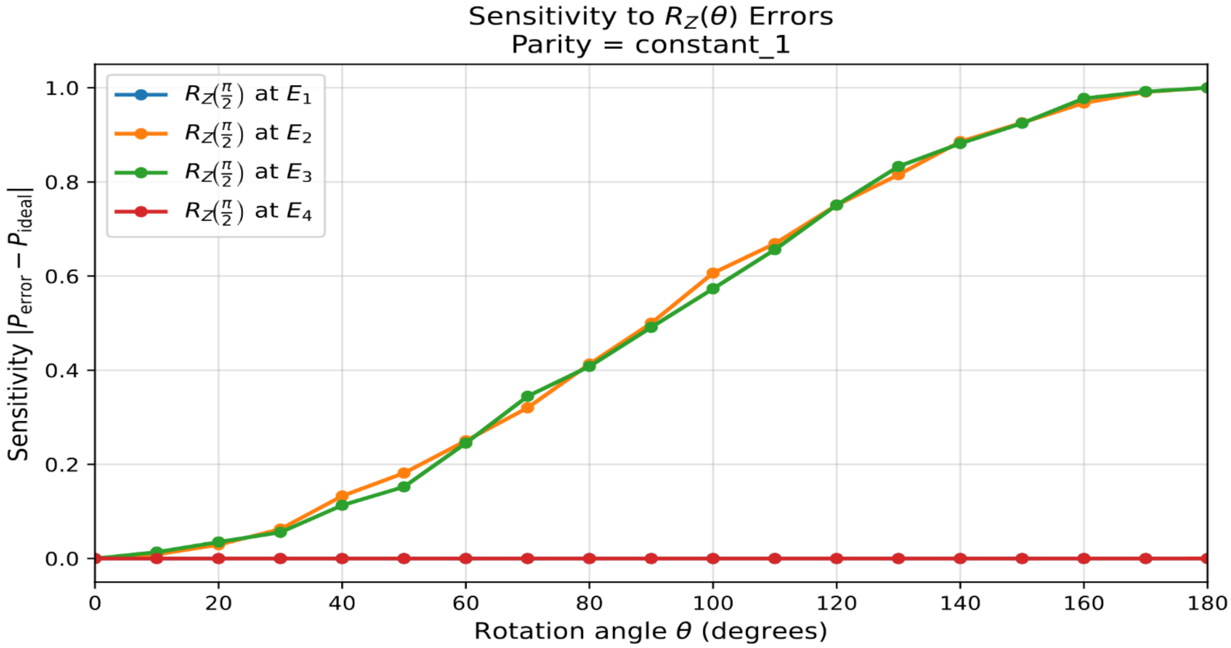

Figure A.3: Sensitivity to $\boldsymbol{R_z(\theta)}$ errors for the Deutsch–Jozsa algorithm.